
Applied Cyber Threat Analysis – Digital Media Analysis Lab

- This lab was conducted within the Virginia Cyber Range Environment.
 - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
 - Cheche Agada. (2024). Digital Media Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University
-

Introduction

One of the major tasks of a digital forensic analyst/examiner/investigator is the examination of digital media for evidence of crime or malicious activity. A typical process starts with the creation of a forensic image of the media (disk drive, flash drive, etc.). The image is a bit-by-bit copy of the original drive. After creating the image, we keep the original drive in a safe place, and work on the image. This prevents knowingly or unknowingly making changes to the original drive – such changes could corrupt the drive and cause any findings made from it to be tossed out as evidence in court. Another reason to create an image is that software (i.e., FTK Imager) used for creating forensic images have built-in capabilities to recover deleted files and show hidden files without additional effort from the examiner.

In this lab, we will conduct preliminary forensic analysis on the image of a flash drive. You will be provided with the image of the flash drive, and you will use the image to complete the following tasks and exercises.

Objective

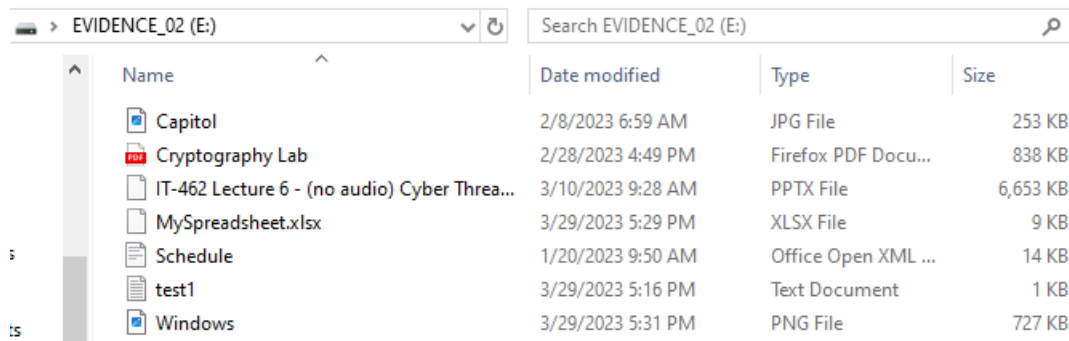
Conduct a forensic analysis of a flash drive.

Task 1 – Download and install FTK Imager and Autopsy on your Windows VM

- FTK Imager (<https://www.exterro.com/ftk-product-downloads/ftk-imager-version-4-7-1>)
- Autopsy (<https://www.autopsy.com/download/>) – Both steps should have been done in week 1.

Task 2 – Drive Content and Chain of Custody

- Take note of the content of the drive



Name	Date modified	Type	Size
Capitol	2/8/2023 6:59 AM	JPG File	253 KB
Cryptography Lab	2/28/2023 4:49 PM	Firefox PDF Docu...	838 KB
IT-462 Lecture 6 - (no audio) Cyber Threa...	3/10/2023 9:28 AM	PPTX File	6,653 KB
MySpreadsheet.xlsx	3/29/2023 5:29 PM	XLSX File	9 KB
Schedule	1/20/2023 9:50 AM	Office Open XML ...	14 KB
test1	3/29/2023 5:16 PM	Text Document	1 KB
Windows	3/29/2023 5:31 PM	PNG File	727 KB

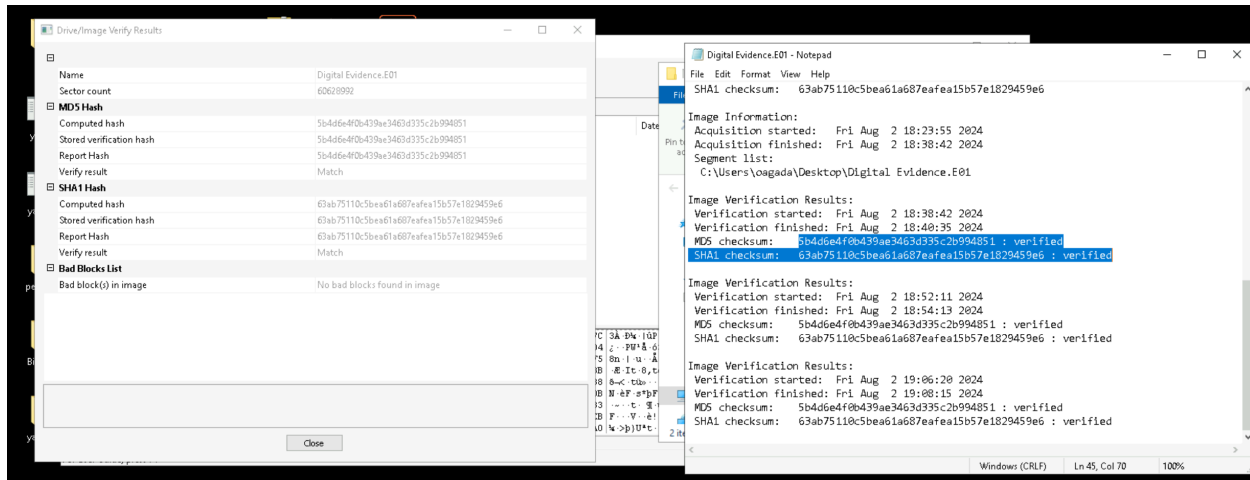
- Create a chain of custody for the drive (see template on Canvas).

NOTE: We are assuming that you have been given the original drive and requested to create the image of the drive and carry out the analysis. The chain of custody (COC) is required to document the movement of the drive throughout the investigation. Complete the COC as best you can, provide information where applicable. Both pages should be filled out, make up the details where necessary.

Task 3 – Verify the image using FTK Imager

- Compare the MD5 and SHA1 hashes in the verification report generated with the ones in the text file downloaded with the image.

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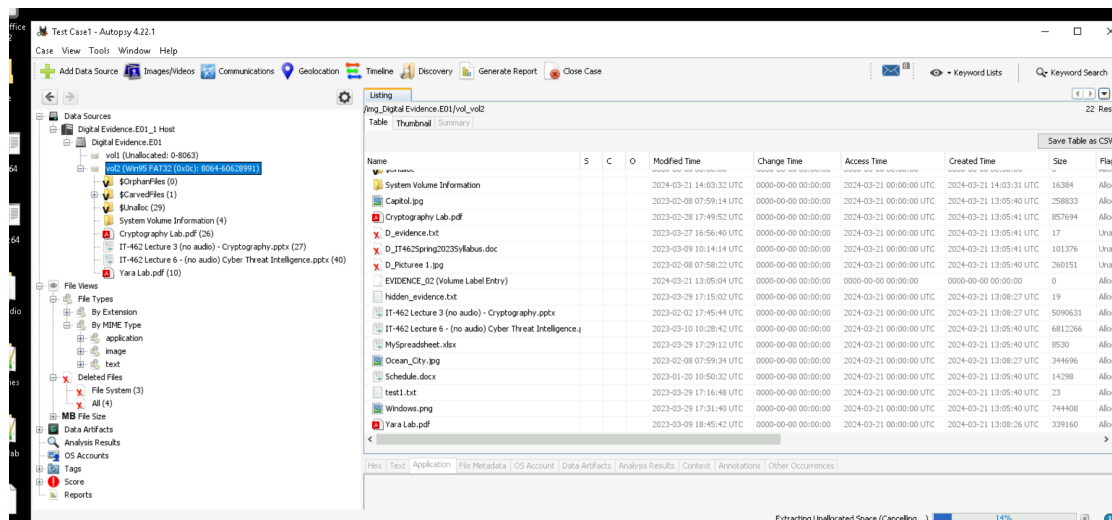
- Are they the same?
 - Yes, the hashes match.

Task 4 – Analyze the image using Autopsy

- Start Autopsy
- Analyze the image and answer the questions that follow.

Class Exercise (Provide screen shots to show your steps)

1. What are the MD5 and SHA1 checksum (hash) of the image?
MD5 checksum: 5b4d6e4f0b439ae3463d335c2b994851
SHA1 checksum: 63ab75110c5bea61a687eafea15b57e1829459e6
2. How many files/folders does the image contain?
There are 16 files



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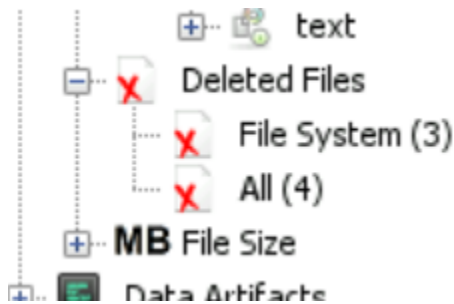
3. How many files/folders are hidden?

There are 2 hidden files



4. How many files/folders are deleted?

There are 4 deleted files



5. Recover the deleted documents and display their contents.

0)

Name	S	C	O	Modified Time	Change Time	Access Time
D_evidence.txt				2023-03-27 16:56:40 UTC	0000-00-00 00:00:00	
D_IT462Spring2023Syllabus.doc				2023-03-09 10:14:14 UTC	0000-00-00 00:00:00	
f0000000.jpg				0000-00-00 00:00:00	0000-00-00 00:00:00	

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100%

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372YX
[~+I
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{K9C
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{RLF
pMQN
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vlt,
aVj}Im,

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nce.pptx (40)

Name	S	C	O	Modified Time	Change Time	Access Time
D_Picturee 1.jpg				2023-02-08 07:58:22 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00
D_evidence.txt				2023-03-27 16:56:40 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00

Error

An error occurred while opening this PDF document. Check the logs for more information. You may continue to use this feature on other PDF documents.

OK

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

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AES Crypt

Start Download (Free)

Proceed To Download (Free)

6. Using the table below, provide the file name, extension, and type for all recovered files.

S/N	Name	Extension	Type	Comments
	Capitol	.jpg	image	none
	Evidence	none	unknown	
	hidden_evidence	.txt	Txt doc	This file is hidden
	Windows	.png	.png image	none
	MySpeadsheet	.xlsx	Excel doc	
	Schedule	.docx	Word doc	
	Yara Lab	.pdf	Pdf document	

TableThumbnailSummary

Name	S	C	O	Modified Time	Change Time
D_IT462Spring2023Syllabus.doc				2023-03-09 10:14:14 UTC	0000-00-00
D_Picture 1.jpg				2023-02-08 07:58:22 UTC	0000-00-00
EVIDENCE_02 (Volume Label Entry)				2024-03-21 13:05:04 UTC	0000-00-00
hidden_evidence.txt				2023-03-29 17:15:02 UTC	0000-00-00
IT-462 Lecture 3 (no audio) - Cryptography.pptx				2023-02-02 17:45:44 UTC	0000-00-00
IT-462 Lecture 6 - (no audio) Cyber Threat Intelligence.pptx				2023-03-10 10:28:42 UTC	0000-00-00
MySpreadsheet.xlsx				2023-03-29 17:29:12 UTC	0000-00-00

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Options

Metadata

Name:

/img_Digital Evidence.E01/vol2/EVIDENCE_02 (Volume Label Entry)

Type:

File System

MIME Type:

application/octet-stream

Size:

0

File Name Allocation:

Allocated

Metadata Allocation:

Allocated

Modified:

2024-03-21 13:05:04 UTC

Accessed:

0000-00-00 00:00:00

Created:

0000-00-00 00:00:00

Changed:

0000-00-00 00:00:00

MD5:

Not calculated

SHA-256:

Not calculated

Hash Lookup Results:

UNKNOWN

Extracting Unallo

TableThumbnailSummary

Name	S	C	O	Modified Time	Change Time
D_IT462Spring2023Syllabus.doc				2023-03-09 10:14:14 UTC	0000-00-00
D_Picture 1.jpg				2023-02-08 07:58:22 UTC	0000-00-00
EVIDENCE_02 (Volume Label Entry)				2024-03-21 13:05:04 UTC	0000-00-00
hidden_evidence.txt				2023-03-29 17:15:02 UTC	0000-00-00
IT-462 Lecture 3 (no audio) - Cryptography.pptx				2023-02-02 17:45:44 UTC	0000-00-00
IT-462 Lecture 6 - (no audio) Cyber Threat Intelligence.pptx				2023-03-10 10:28:42 UTC	0000-00-00
MySpreadsheet.xlsx				2023-03-29 17:29:12 UTC	0000-00-00

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Options

Strings

Extracted Text

Translation

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Troy Fracyon
6/29/2025

TableThumbnailSummary

Name	S	C	O	Modified Time	Change Time
IT-462 Lecture 6 - (no audio) Cyber Threat Intelligence.i				2023-03-10 10:28:42 UTC	0000-00-00 00:00:00
MySpreadsheet.xlsx				2023-03-29 17:29:12 UTC	0000-00-00 00:00:00
Ocean_City.jpg				2023-02-08 07:59:34 UTC	0000-00-00 00:00:00
Schedule.docx				2023-01-20 10:50:32 UTC	0000-00-00 00:00:00
test1.txt				2023-03-29 17:16:48 UTC	0000-00-00 00:00:00
Windows.png				2023-03-29 17:31:40 UTC	0000-00-00 00:00:00
Yara Lab.pdf				2023-03-09 18:45:42 UTC	0000-00-00 00:00:00

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

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Reset

Extracting Unallocated Space

TableThumbnailSummary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
IT-462 Lecture 6 - (no audio) Cyber Threat Intelligence.i				2023-03-10 10:28:42 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:05:4
MySpreadsheet.xlsx				2023-03-29 17:29:12 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:05:4
Ocean_City.jpg				2023-02-08 07:59:34 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:08:2
Schedule.docx				2023-01-20 10:50:32 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:05:4
test1.txt				2023-03-29 17:16:48 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:05:4
Windows.png				2023-03-29 17:31:40 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:05:4
Yara Lab.pdf				2023-03-09 18:45:42 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024-03-21 13:08:2

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

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Page 1 / 5

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TableThumbnailSummary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
System Volume Information				2024-03-21 14:03:32 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
Captrol.jpg				2023-02-08 07:59:14 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
Cryptography Lab.pdf				2023-02-28 17:49:52 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
D_evidence.txt				2023-03-27 16:56:40 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
D_IT462Spring2023syllabus.doc				2023-03-09 10:14:14 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
D_Picture 1.jpg				2023-02-08 07:58:22 UTC	0000-00-00 00:00:00	2024-03-21 00:00:00 UTC	2024
EVIDENCE_02 (Volume Label Entry)				2024-03-21 13:05:04 UTC	0000-00-00 00:00:00	0000-00-00 00:00:00	0000

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

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Reset

Extracting Unallocated Space (Cancelling...)

TableThumbnailSummary

Name	S	C	O	Modified Time
IT-462 Lecture 6 - (no audio) Cyber Threat Intelligence.i				2023-03-10 10:28:42
MySpreadsheet.xlsx				2023-03-29 17:29:12
Ocean_City.jpg				2023-02-08 07:59:34
Schedule.docx				2023-01-20 10:50:32
test1.txt				2023-03-29 17:16:48
Windows.png				2023-03-29 17:31:40
Yara Lab.pdf				2023-03-09 18:45:42

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

StringsExtracted TextTranslation

Page: 1 of 1 PageMatches on page: - of - Match100%

Sheet1

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